Comprehensive Requirements Catalog — WariMitra

Document ID: WM-REQ-005

Classification: Full Lifecycle Requirements Specification & Taxonomy

Standard: IEEE Std 830-1998 / BABOK v3 Chapter 6 (Requirements Lifecycle Management)

Target Audience: Lead Business Analysts, Technical Architects, QA Leads, Interview Evaluation Panels

Evidence Sources: `functions/src/index.ts`, `ARCHITECTURE.md`, `docs/BRD.md`, `docs/FRD.md`, `mobile/lib/`, `frontend/src/`

1. Requirements Taxonomy & ID Nomenclature

Prefix	Requirement Category	Business Scope
BR-xxx	Business Requirements	High-level organizational goals, legal mandates, and operational objectives.
FR-xxx	Functional Requirements	Specific system behaviors, capabilities, inputs, outputs, and user interactions.
NFR-xxx	Non-Functional Requirements	Performance, latency, scalability, reliability, and accessibility benchmarks.
DR-xxx	Data Requirements	Entity structures, schema constraints, retention rules, and indexing.
INT-xxx	Integration Requirements	External third-party integrations (Google Maps, FCM, Web3 testnet, Telco).
SEC-xxx	Security Requirements	Authentication, RBAC, encryption, geofencing, rate limiting, and audit trails.
REP-xxx	Reporting & Analytics Requirements	Dashboards, singleton aggregators, telemetry, and administrative metrics.

2. Business Requirements (BR)

Req ID	Requirement Statement	Business Objective	Evidence in Repo	Module	Priority	Status
BR-001	The system shall provide an instantaneous, zero-confusion emergency distress channel for pilgrims walking along the pilgrimage route.	Reduce fatal response times during medical crises and stampedes.	`docs/BRD.md:16`, `README.md:26`	Emergency SOS	Must Have	**Implemented**
BR-002	The system shall enable geospatial discovery of active medical camps and available beds within a walking radius.	Prevent overloading civil hospitals and reduce distance to first aid.	`docs/BRD.md:18`, `functions/src/index.ts:122`	Healthcare	Must Have	**Implemented**
BR-003	The system shall provide government	Eliminate stampedes and mitigate	`docs/BRD.md:19`, `docs/FRD.md:42`	Crowd Control	Must Have	**Implemented**

	government	and mitigate				

Req ID	Requirement Statement	Business Objective	Evidence in Repo	Module	Priority	Status
	and police authorities with real-time situational visibility over crowd density and queue wait times.	extreme crowd crush risks.				
BR-004	The system shall establish an unalterable, non-repudiable audit log of emergency responses for judicial and administrative transparency.	Protect administration from false liability claims and prove response diligence.	`docs/WEB3_ARCHITECTURE.md:3`, `functions/src/index.ts:237`	Trust Layer	Should Have	**Implemented**
BR-005	The system shall maintain operational stability at zero recurring	Satisfy state public finance constraints prohibiting cryptocurrency	`docs/WEB3_ARCHITECTURE.md:26-31`	Trust Layer	Must Have	**Implemented**

	recurring blockchain gas cost.	cy expenditure.				

3. Functional Requirements (FR)

Req ID	Requirement Statement	Module	Priority	Dependencies	Evidence Path	Acceptance Criteria	Status
FR-001	The mobile client shall require a continuous 3-second hold gesture on the SOS button to initiate distress dispatch, providing visual progress and haptic vibration.	Emergency SOS	Must Have	Device Sensors	`mobile/lib/screens/sos_screen.dart`, `VARIMITRA_SPEAKER_CONTEXT.md:141`	Given user is on SOS screen, When button is held for 3.0s, Then circular progress completes and API is invoked; if released early, trigger cancels.	**Implemented**
FR-002	The backend	Emergency SOS	Must Have	FR-001	`functions/src/`	Given an identical	**Implemented**

	backend				`s/src/`	identical	

Req ID	Requirement Statement	Module	Priority	Dependencies	Evidence Path	Acceptance Criteria	Status
	shall enforce idempotency on SOS creation using a client-generated UUID, returning the existing record if retransmitted.				`index.ts:43-56`	`idempotencyKey`, When submitted within 60s, Then system returns existing `alertId` with no duplicate database write.	
FR-003	The system shall rate-limit emergency SOS creation to a maximum of 5 alerts per minute per user account.	Emergency SOS	Must Have	Auth	`functions/src/index.ts:59-70`	Given a user has created 5 alerts in the last 60s, When 6th alert is attempted, Then function aborts with `resource-exhausted`	**Implemented**

Req ID	Requirement Statement	Module	Priority	Dependencies	Evidence Path	Acceptance Criteria	Status
						`exhausted` .	
FR-004	The system shall validate that SOS coordinates fall within the geographical bounding box of India (Lat: 8.0–35.0, Lon: 68.0–97.5).	Emergency SOS	Must Have	GPS Telemetry	`functions/src/index.ts:72-80`	Given coordinates outside India bounds, When SOS is submitted, Then function rejects request with `failed-precondition`.	**Implemented**
FR-005	The system shall broadcast FCM push notifications to all authenticated users with roles `police_officer`,	Notification Engine	Must Have	FR-001, RBAC	`functions/src/index.ts:182-218`	Given a new document in `/sosAlerts/`, When `onSOSCreate` fires, Then multicast notification is	**Implemented**

Req ID	Requirement Statement	Module	Priority	Dependencies	Evidence Path	Acceptance Criteria	Status
	`ficer`,					n is	
	`medical_officer`, and `admin` upon SOS creation.					dispatched to all active responder tokens.	
FR-006	The system shall calculate Geohash bounding boxes to query active medical camps within a user-specified radius (default 10 km) sorted by Haversine distance.	Medical Services	Must Have	Location Services	`functions/src/index.ts:122-177`	Given valid lat/lon, When `getMedicalCamps` is invoked, Then returns active camps sorted ascending by `distanceKm`.	**Implemented**
FR-007	The mobile client shall allow	Family Locator	Should Have	Auth, GPS	`ARCHITECTURE.md:219-224`, `mobile/`	Given a valid 6-char code, When user	**Implemented**

Req ID	Requirement Statement	Module	Priority	Dependencies	Evidence Path	Acceptance Criteria	Status
	allow users to generate or join a family tracking group using a unique 6-character alphanumeric invite code.				`mobile/lib/screens/family_locator_screen.dart`	When user enters code, Then user's UID is appended to `memberIds` array in `familyGroups/{groupId}`.	
FR-008	The mobile app shall stream device GPS coordinates and battery percentage to `users/{userId}/liveLocation/current` every 30 seconds.	Pilgrim Tracking	Should Have	Location Services	`mobile/lib/services/location_service.dart`, `VARIMITRA_SPEAKER_CONTEXT.md:155`	Given active pilgrimage mode, When 30s timer elapses, Then current coordinates and battery level are written to Firestore.	**Implemented**

Req ID	Requirement Statement	Module	Priority	Dependencies	Evidence Path	Acceptance Criteria	Status
FR-009	The system shall allow pilgrims and authorities to submit missing person reports including name, age, physical description, last-seen GPS, and photo upload.	Missing Persons	Must Have	Firebase Storage	`firestore.rules:111-117`, `mobile/lib/screens/missing_person_screen.dart`	Given report data and image, When submitted, Then photo is saved to Cloud Storage and metadata document created in `missingPersons`.	**Implemented**
FR-010	The command center shall render real-time Darshan queue status for temple gates,	Temple Ops	Must Have	Firestore Listeners	`frontend/src/app/temple/`, `ARCHITECTURE.md:225-230`	Given active temple queues, When Firestore listener receives snapshot, Then UI displays	**Implemented**

Req ID	Requirement Statement	Module	Priority	Dependencies	Evidence Path	Acceptance Criteria	Status
	gates, showing queue headcount and estimated wait times.					displays wait time in minutes and gate status.	
FR-011	The command center shall provide a virtualized incident feed rendering live SOS alerts with audio-visual indicators for high/critical severity.	Command Center	Must Have	React-Window	`frontend/src/app/sos/`, `frontend/package.json:33`	Given >100 concurrent alerts, When rendered in SOS feed, Then `react-window` ensures zero frame drops or DOM memory leaks.	**Implemented**
FR-012	The system shall display civic sanitation	Sanitation	Could Have	Firebase Storage	`firestore.rules:97-110`, `frontend/src/app/`	Given citizen waste report with photo, When	**Implemented**

Req ID	Requirement Statement	Module	Priority	Dependencies	Evidence Path	Acceptance Criteria	Status
	sanitation facilities and allow crowdsourced reporting of accumulated solid waste with photo proof.				`app/sanitation/`	When created, Then report appears in municipal dashboard for sanitation crew dispatch.	

4. Non-Functional Requirements (NFR)

Req ID	Requirement Statement	Category	Target Metric	Measured / Verified	Evidence Path
NFR-001	Emergency SOS end-to-end creation latency from client trigger to database persistence shall be under 2 seconds.	Performance / Latency	< 2.0 seconds	**1.2s (p95)**	`README.md:243`, `ARCHITECTURE.md:428`
NFR-002	Cloud Firestore database read	Performance / Latency	< 500 ms	**320ms (p95)**	`README.md:244`, `ARCHITECTURE.`

Req ID	Requirement Statement	Category	Target Metric	Measured / Verified	Evidence Path
	database read queries for operational views shall execute in under 500 milliseconds.				ARCHITECTURE.md:429
NFR-003	Real-time state synchronization via Firestore listeners shall propagate to active command clients in under 2 seconds.	Real-Time Sync	< 2.0 seconds	**1.5s (p95)**	README.md:245, ARCHITECTURE.md:430
NFR-004	Photo file uploads (up to 5MB) to Firebase Storage shall complete in under 5 seconds under standard 4G connectivity.	Throughput	< 5.0 seconds	**3.8s (p95)**	README.md:246, ARCHITECTURE.md:431
NFR-005	The backend architecture shall support at least 10,000 concurrent connected users	Scalability	10,000+ CCU	Verified by Serverless Architecture	README.md:249-250, ARCHITECTURE.md:437

Req ID	Requirement Statement	Category	Target Metric	Measured / Verified	Evidence Path
	connected users and 100+ peak emergency SOS alerts per hour.				
NFR-006	Mobile user interface shall support native Marathi (`mr`), Hindi (`hi`), and English (`en`), with default routing to Marathi.	Accessibility / Localization	100% UI String Localization	Implemented via `language_provider.dart`	`ARCHITECTURE.md:115`, `mobile/lib/screens/home_screen.dart`
NFR-007	Client applications shall implement offline persistence to allow reading cached maps and queues during intermittent cellular disconnects.	Reliability / Fault Tolerance	Zero crash on connection loss	Local Firestore cache enabled	`ARCHITECTURE.md:111,468`

5. Security & Access Control Requirements (SEC)

Req ID	Requirement Statement	Enforcement Level	Evidence Path
SEC-001	User authentication shall support phone-based OTP (SMS) and email/password via Firebase Authentication.	Identity Provider	`ARCHITECTURE.md:341-344`
SEC-002	System authorization shall enforce Role-Based Access Control (RBAC) across 7 distinct roles using Firebase Custom Claims (`token.userType`).	Identity & Auth	`ARCHITECTURE.md:345-366`, `firestore.rules:10-29`
SEC-003	Database security rules shall prevent non-authorities from modifying or updating emergency alerts, medical camp records, or police patrols.	Database Level	`firestore.rules:59-85`
SEC-004	System audit logs stored in `auditLogs` shall be append-only and strictly restricted to Cloud Function writes (`allow write: if false;`).	Immutability / Compliance	`firestore.rules:174-177`
SEC-005	Cloud Storage file access shall enforce strict ownership boundaries for personal missing persons uploads.	Storage Rules	`storage.rules:318-335`

6. Integration Requirements (INT)

Req ID	External System / Interface	Interface Method	Protocol / Format	Purpose
INT-001	Google Maps Platform API	Client SDK / JS Loader	HTTPS / GeoJSON	Rendering spatial maps, cluster markers, and directions routing.
INT-002	Firebase Cloud Messaging (FCM)	Admin SDK (`sendEachForMulticast`)	HTTP/2 / JSON	Multicast dispatch of emergency alerts to responder devices.
INT-003	Ethereum Sepolia Testnet	Ethers.js v6 JSON-RPC	Web3 RPC / 0-ETH Tx	Anchoring SHA-256 canonical event hashes into blockchain block data.
INT-004	State 108 Emergency Medical Services	Hardware Telco Click-to-Dial	Telephony `tel:` URI	Direct dialing escalation to government ambulance dispatchers.

7. Data & Reporting Requirements (DR / REP)

- **DR-001 (Spatial Indexing):** All geographic entities (`sosAlerts`, `medicalCamps`, `communityReports`) must store a 9-character Geohash alongside `latitude` and `longitude` (`functions/src/index.ts:89,147`).
- **DR-002 (Idempotency Key Uniqueness):** The `idempotencyKey` field in `sosAlerts` must be indexed and checked prior to document insertion (`functions/src/index.ts:43-56`).
- **REP-001 (Scheduled Aggregations):** A scheduled Cloud Function (`updateDashboardMetrics`) shall execute every 1 minute via Cloud Pub/Sub to aggregate pending SOS count, active medical camps, and registered pilgrims into `analytics/dashboardMetrics` (`functions/src/index.ts:318-335`).